

Tulips Project: Suitable hardware

The tulips monitoring system performs monitoring operations. This stage involves identifying devices that can effectively sense environmental conditions, process data, display real-time information, and control external actuators. Each component has been chosen based on its compatibility with the microcontroller as well as its functionality

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Select suitable hardware as follows:

- a) ESP32 DevKit V1 microcontroller
- b) DHT22 temperature and humidity sensor
- c) MQ-5 Gas sensor
- d) A 1.3-inch I2C OLED display
- e) 5V relay module

ESP32 DevKit V1 microcontroller

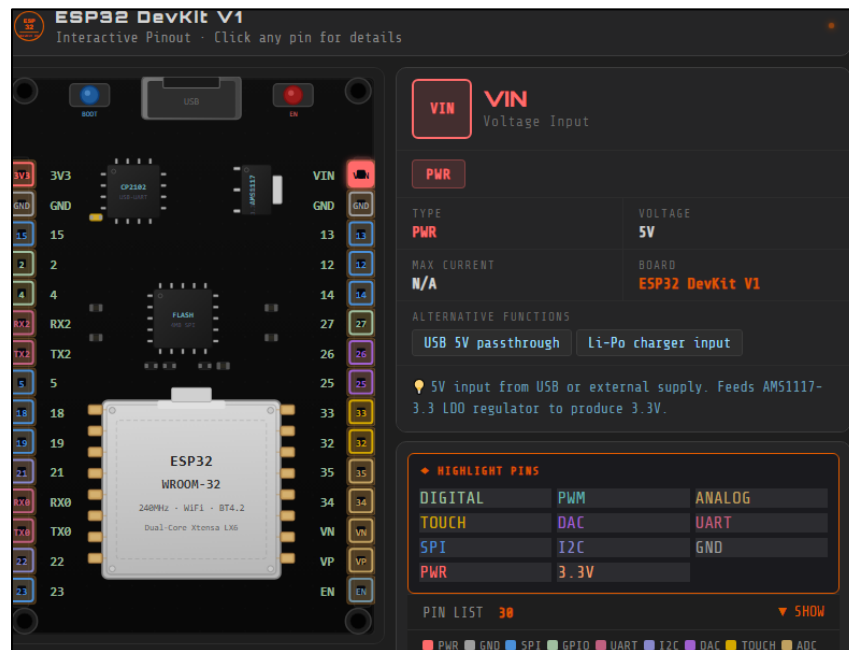


Figure 1: Esp32 DevKit v1 microvontroller

The ESP32-DevKit v1 is a widely recognized development board built around Espressif's ESP32-WROOM-32 ESP32 module. This powerful chip integrates Wi-Fi and Bluetooth connectivity, making it an ideal choice for a vast array of Internet of Things (IoT) Applications (Espressif Systems, 2020). It is also selected due to its ability to handle multiple sensor inputs

DHT22 temperature and humidity sensor

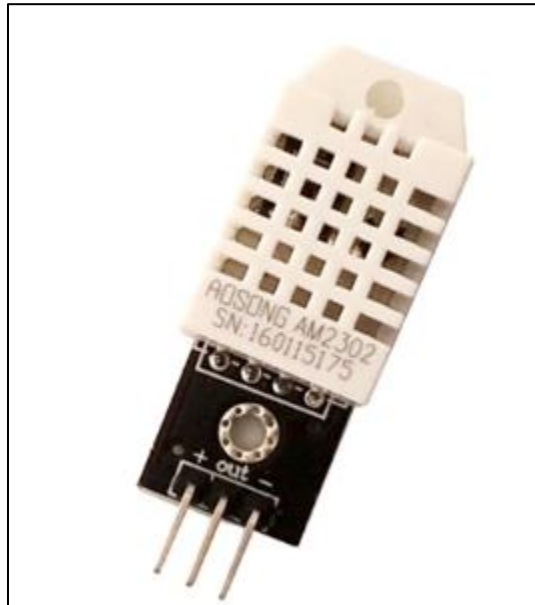


Figure 2: DHT22 temperature and humidity sensor

The **DHT22 AM2302** was selected for measuring environmental temperature and humidity. It is a low-cost digital sensor that uses a capacitive humidity sensing element and a thermistor to measure surrounding air conditions (Aosong Electronics, 2018)

MQ-5 Gas sensor



Figure 3: MQ-5 Gas sensor

The MQ-5 gas sensor uses 5nO^2 semiconductor material whose conductivity changes in the presence of flammable gases (Winsen Electronics, 2018).

1.3-inch I2C OLED display

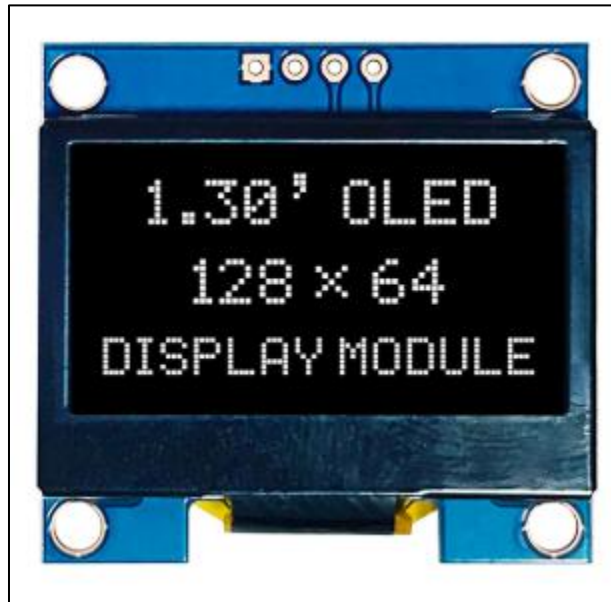


Figure 4: 1.3-inch I2C OLED display

A **1.3-inch OLED display I2C** is a compact visual output interface that displays real-time readings. It offers a resolution of 128×64 pixels and communicates via the I2C protocol, making it efficient for embedded systems with limited GPIO pins (Solomon Systech, 2017). It was selected due to its low power consumption and high readability.

5V relay module

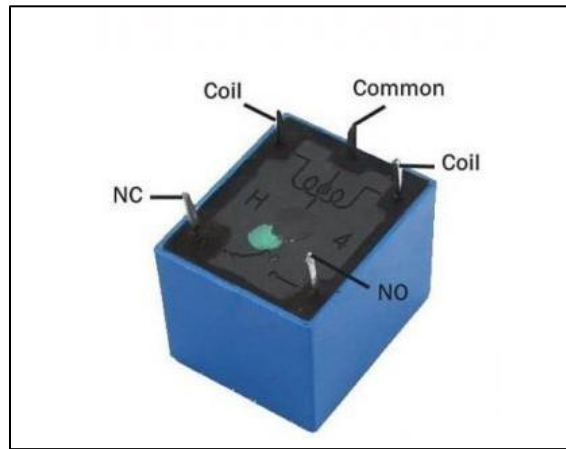


Figure 5: 5V relay module

The 5V relay is an automatic switch commonly used in automatic control circuits to control high current with a low-current signal. It allows the ESP32 to safely control devices such as fans, pumps, or alarms without direct exposure to high current circuits (Songle Relay Datasheet, 2019).

References

- Espressif Systems. (2020). ESP32-WROOM-32 Datasheet*
- Aosong Electronics. (2018). DHT22/AM2302 Datasheet*
- Zhengzhou Winsen Electronics Technology Co., Ltd. (2018). MQ-5 Gas Sensor Datasheet*
- Solomon Systech. (2017). SSD1306/SH1106 OLED Datasheet*
- Songle Relay. (2019). SRD-05VDC Relay Module Datasheet*